

DR. TINGNAN BAO

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IEEE Senior Member • CIE Young Scientist

EDUCATION

Doctor of Philosophy in Electrical and Computer Engineering

University of Victoria

05/2016 - 10/2020

Victoria, Canada

Master of Science in Information Technology

KTH Royal Institute of Technology

09/2013 - 10/2015

Stockholm, Sweden

Bachelor of Science in Electrical Engineering

Liaoning Technical University

09/2003 - 07/2007

Fuxin, China

PROFESSIONAL EXPERIENCE

Research Associate

University of Ottawa

06/2026 - Present

Ottawa, Canada

- Conducted research on quantum algorithms over networking.
- Collaborated on the Mitacs Accelerate project with Ericsson Canada on federated learning for wireless networks.

Research Fellow

University of Ottawa

06/2023 - 05/2026

Ottawa, Canada

- Conducted research in AI and machine learning applications for 6G technologies, focusing on task offloading for UAV-assisted secure communications in smart farming environments.
- Conducted research on deep reinforcement learning (DRL) for enhancing the security of satellite communications.
- Designed and implemented an agentic multi-agent DRL strategy to optimize energy efficiency in communication systems.
- Engaged in collaboration, proposal writing, and regular technical coordination meetings with project partners.
- Applied MATLAB and Python extensively for conducting statistical analysis and optimizing system performance.

Advisory Researcher

Lenovo Research

12/2020 - 04/2023

Beijing, China

- Conducted research on future 6G technologies, including reconfigurable intelligent surfaces (RIS) and AI for physical layer (PHY), with a focus on patent layouts and standardization efforts.
- Contributed to standardization efforts on AI for PHY in 3GPP RAN.
- Engaged in industry organizations and forums such as 3GPP, IMT-2030 (6G), and the Future Forum for 6G.
- Provided supervision and coaching to RAN researchers, fostering a collaborative and productive research environment.

Graduate Researcher

University of Victoria

05/2016 - 10/2020

Victoria, Canada

- Developed and implemented a deep learning solution using a deep neural network to predict the secrecy outage probability in UAV-based transmission and verified by simulation works.
- Proposed and implemented an analytical framework for evaluating secrecy performance in UAV-based transmission.
- Conducted comprehensive research on ground-based networks with multiple antenna and massive MIMO transmission, focusing on the secrecy performance of random unitary beamforming in emerging wireless transmission systems.

5G Algorithm Researcher Intern

Create-Net Research Center

03/2015 - 10/2015

Trento, Italy

- Investigated dynamic resource allocation in heterogeneous mobile networks (LTE and 5G network).
- Proposed dynamic strict fractional frequency reuse method to relieve inter-cell interference in different cell sectors.
- Collaborated with a Ph.D., a future networks lab in UNITN, and researchers from this research centre.

Technical Team Manager / Telecommunication Engineer

Ericsson China (Southern, Western, and Central Regions)

03/2008 - 01/2013

Multiple Cities, China

- Managed deployment, optimization, and performance evaluation of Ericsson RAN products, including W-CDMA RNC (30+) and RBS (6000+) nodes.
- Supervised a local engineering team of 40-50 people across 12 cities, closely coordinating with global service delivery center.

RESEARCH FUNDING

Core Member & Research Intern, Mitacs Accelerate with Ericsson	02/2025 - Present
Core Member, NSERC Canada Research Chairs Program	06/2024 - 01/2025
Core Member & Research Intern, Mitacs Accelerate with Nokia Bell Labs	05/2023 - 06/2024
Principal Investigator, Beijing Research Foundation	01/2021 - 12/2022
Core Member, Qatar National Research Fund (NPRP13S-0130-200200)	01/2021 - 12/2022
Core Member, National Natural Science Foundation of China (No. 62001381)	01/2021 - 12/2023
Core Member, NSERC Discovery Grant (No. 5001-2016)	01/2016 - 12/2020
Core Member, Qatar University Grant (No. QUSM-CENG-2018-1)	01/2018 - 12/2020
Core Member, Ooredoo Research Grant	01/2016 - 12/2018

TEACHING EXPERIENCE

Teaching Assistant , University of Victoria	05/2016 - 07/2020
- ECE 300 - Linear Circuits II	Spring 2020
- ECE 516 - Advanced Wireless Communications	Fall 2019
- ECE 310 - Digital Signal Processing I	Spring 2019
- ECE 456 - Wireless and Mobile Communications	Summer 2018
- ECE 350 - Communication Theory & Systems: I	Fall 2018, Fall 2017, Fall 2016
Research Mentorship , University of Ottawa	06/2023 – Present
- S.F. Mobarak (Postdoc): RIS-assisted mmWave ISAC → IEEE PIMRC 2025, IEEE ICC 2025 & 2026	
- L. Nobrega (M.Sc.): DRL for smart agriculture → IEEE PIMRC 2024, IEEE ICMLCN 2024	
Research Intern Mentor , Motorola Mobility	06/2021 – 08/2021
- Mentored interns from Southeast University on 3GPP standardization	

ACADEMIC LEADERSHIP & COLLABORATION

Workshop Chair: IoTIR 2026 (Tangshan, China)
Technical Program Committee: IEEE GLOBECOM (2025, 2026), IEEE ICC (2024, 2025, 2026), IEEE PIMRC (2024, 2025), IEEE PacRim 2024, IUI 2022
Academia and Industry Partnerships: Mitacs Accelerate with Ericsson and Nokia Bell Labs
Standards & Forums: 3GPP RAN delegate (2021 to 2023); IMT-2030 (6G) Promotion Group participant
Journal Reviewer: 100+ manuscripts for IEEE journals
Invited Talks: ETIS Laboratory, CY Cergy Paris Université, France (2025); Shenzhen University, China (2022)

ENTREPRENEURSHIP & LEADERSHIP

President, EIT ICT Labs Alumni CSSA, Trento, Italy	09/2014 - 10/2015
Project Leader, EIT ICT Summer School “Smart Cities”, Nice, France	06/2014 - 07/2014
Co-founder, Mofit Health Solutions, Stockholm, Sweden	01/2014 - 07/2014
Student Representative, EIT Innovation Forum INNOVEIT 2015, Budapest, Hungary	05/2015

HONORS AND AWARDS

- i2I Skills Training Grant by Natural Sciences and Engineering Research Council of Canada	2026
- Exemplary Reviewer in IEEE Transaction on Network Science and Engineering by IEEE Communications Society	2025
- Global 50 of 2023 Chinese Government Award for Outstanding Self-financed Student Abroad	2024
- Best Paper Award at IEEE PacRim	2024, 2019
- Beijing Research Foundation	2022
- University of Victoria Research Assistantship and Fellowship	2016-2020
- University of Victoria Graduate Award	2016
- Master Scholarship by European Institute of Innovation and Technology	2013
- First-class Graduate Student Scholarship at Liaoning Technical University	2004-2007
- Liaoning Provincial Scholarship (1-st), China	2004

PUBLICATIONS

Summary: 13 journal articles, 13 conference papers, 13 patents, 5 manuscripts under review

- Journal Articles Under Review

- [1]. **T. Bao**, M. Erol-Kantarci, "Task-Specific Hybrid Reinforcement Learning for RSMA-based RIS-assisted Secure Satellite Systems." *IEEE Trans. Mach. Learn. Commun. Netw.*, Submitted, 2025. (Minor Revision)
- [2]. **T. Bao**, M. Elsayed, et al., "Deep Reinforcement Learning with Self-distillation for RIS-assisted Secure Resource-constrained Wireless Networks." *IEEE Trans. Veh. Technol.*, Submitted, 2026.
- [3]. **T. Bao**, M. Elsayed, et al., "Agentic Quantum Deep Reinforcement Learning for RAN Slicing." *IEEE Trans. Netw. Sci. Eng.*, Submitted, 2026.
- [4]. **T. Bao**, M. Elsayed, et al., "Agentic Multi-Agent DRL for Joint Beamforming and Phase Shift Optimization in Multi-RIS Secure Communications." *IEEE Trans. Cogn. Commun. Netw.*, Submitted, 2026.
- [5]. **T. Bao**, M. Elsayed, et al., "Generative AI for RIS-Assisted Secure Non-Terrestrial Networks." *IEEE Veh. Technol. Mag.*, Submitted, 2026.

- Refereed Journal Articles

- [1]. **T. Bao**, M. Elsayed, P. Iturria-Rivera, Y. Ozcan, M. Bavand, M. Erol-Kantarci, "Sum Secrecy Rate Maximization for Cascaded Far-to-Near-Field Communications." *IEEE Wirel. Commun. Lett.*, pp. 1-5, Jun., early access, 2026.
- [2]. **T. Bao**, M. Elsayed, P. Iturria-Rivera, Y. Ozcan, M. Bavand, M. Erol-Kantarci, "Intelligent Discrete RIS Phase Configuration in Secure Non-Terrestrial Networks: A Heuristic Conditional GAN Approach," *IEEE Trans. Netw. Sci. Eng.*, vol.13, pp. 9339 - 9357, May, 2026.
- [3]. F.Xu, Y Wang, T. A. Gulliver, Y Xie, C. Wang, R.Jiang, **T. Bao**, E. G. Lim, Z. Lin, R. Samy. "Finite Blocklength Relaying Communication With Unitary Beamforming and Energy Harvesting: Fairness-Oriented Design." *IEEE Trans. Ind. Inform.*, vol. 22, no. 1, pp. 463 - 473, Jan., 2026.
- [4]. **T. Bao**, A. Syed, WS. Kennedy, M. Erol-Kantarci. "Sustainable Task Offloading in Secure UAV-assisted Smart Farm Networks: A Multi-Agent DRL with Action Mask Approach." *IEEE Trans. Netw. Serv. Manag.*, vol. 22, no.4, pp. 3191-3200, Aug., 2025.
- [5]. F. Xu , B. Duo, Y. Xie, G. Pan, Y. Yang, Y. Ye, **T. Bao**, T. A Gulliver, L. Zhang. "Multi-UAV assisted Mixed FSO/RF Communication Network for Urgent Tasks: Fairness Oriented Design with DRL." *IEEE Trans. Veh. Technol.*, vol. 74, pp. 1736-1741, Sep., 2024.
- [6]. A. Termehchi, **T. Bao**, A. Syed, WS. Kennedy, M. Erol-Kantarci. "Goal-oriented Reinforcement Learning in THz-enabled UAV-aided Network Using Supervised Learning." *IEEE Open J. Commun. Soc.*, vol. 5, pp. 5027-5036, Aug., 2024.
- [7]. **T. Bao**, A. Syed, WS. Kennedy, M. Erol-Kantarci. "SDOR Analysis for Task Offloading in Smart Farm Secure UAV-Assisted Communication Systems." *IEEE Trans. Veh. Technol.*, vol. 73, no. 11, pp. 17776-17781, Jun., 2024.
- [8]. **T. Bao**, H. Wang, W.-J. Wang, H.-C. Yang, and M. O. Hasna. "Secrecy Outage Performance Analysis of UAV-assisted Relay Communication Systems with Multiple Aerial and Ground Eavesdroppers." *IEEE Trans. Aerosp. Electron. Syst.*, vol. 58, no. 3, pp. 2592-2600, 2022.
- [9]. **T. Bao**, J. Zhu, H.-C. Yang, and M. O. Hasna. "Secrecy Outage Performance of Ground-to-Air Communications with Multiple Aerial Eavesdroppers and its Deep Learning Evaluation." *IEEE Wirel. Commun. Lett.*, vol. 9, no. 9, pp. 1351-1355, Sep. 2020.
- [10]. **T. Bao**, H.-C. Yang, and M. O. Hasna. "Secrecy Outage Analysis over Random Beamforming Transmission with User Scheduling: Collusion vs. Exclusion." *IEEE Signal Process. Lett.*, vol. 27, no. 1, pp. 1295-1299, Jul. 2020.
- [11]. **T. Bao**, H.-C. Yang, and M. O. Hasna. "Secrecy Performance Analysis of UAV-Assisted Relaying Communication Systems." *IEEE Trans. Veh. Technol.*, vol. 69, no. 1, pp. 1122-1126, Nov. 2019.
- [12]. H.-C. Yang, **T. Bao**, and S. Alouini, "Transient Performance Limits for Wireless Transmissions over Fading Channels." *IEEE Trans. Veh. Technol.*, vol. 69, no. 11, pp. 13970-13973, Nov. 2020.
- [13]. **T. Bao**, H.-C. Yang, and M. O. Hasna. "Physical layer secrecy performance of multiple antennas transmission with partial legitimate user CSI." *IET Commun.*, vol. 13, no. 15, pp. 2285-2295, Aug. 2019.

- Refereed Conference Papers

- [1]. S.F. Mobarak, **T. Bao**, M. Erol-Kantarci. "Secrecy-Driven ISAC Optimization in mmWave via Twin Delayed DDPG and RIS-Assisted Beamforming." *IEEE ICC*, Glasgow, Scotland, UK, May., 2026, pp. 1-6.
- [2]. S.F. Mobarak, **T. Bao**, M. Erol-Kantarci. "CRB Minimization Using Twin Delayed DDPG for Semi-Self Sensing Ac-

tive RIS-Assisted mmWave ISAC.” *IEEE PIMRC*, Istanbul, Türkiye, Sep., 2025, pp. 1-6.

- [3]. **T. Bao**, M. Erol-Kantarici. “Heuristic Deep Reinforcement Learning for Phase Shift Optimization in RIS-assisted Secure Satellite Communication Systems with RSMA.” *IEEE ICC*, Montreal, CA, Jun., 2025, pp. 1-6.
- [4]. S.F. Mobarak, **T. Bao**, M. Erol-Kantarici. “Sum Rate Enhancement using Machine Learning for Semi-Self Sensing Hybrid RIS-Enabled ISAC in THz Bands.” *IEEE ICC*, Montreal, CA, Jun., 2025, pp. 1-6.
- [5]. L. Nobrega, A. Termehchi, **T. Bao**, A. Syed, S. Kennedy, M. Erol-Kantarici. “Hierarchical Deep Reinforcement Learning With Information Freshness in Smart Agriculture Applications.” *IEEE PIMRC*, Valencia, ES, Sep., 2024, pp. 1-6.
- [6]. F. Wang, R. Jiang, Y. Liu, **T. Bao**, M. Peng. “On the Outage of Far-Reach Communications via RIS-Empowered Satellites.” *IEEE PacRim*, Victoria, CA, Aug., 2024, pp. 1-6. (**Best Paper Award**)
- [7]. L. Nobrega, A. Termehchi, **T. Bao**, A. Syed, S. Kennedy, M. Erol-Kantarici. “AoI-Aware Trajectory Planning for Smart Agriculture Using Proximal Policy Optimization.” *IEEE ICMLCN*, Stockholm, SE, May, 2024, pp. 1-6.
- [8]. **T. Bao**, H. Wang, W.-J. Wang, H.-C. Yang, and M. O. Hasna. “Performance Analysis of RIS-aided Communication Systems over the Sum of Cascaded Rician Fading with imperfect CSI.” *IEEE WCNC*, Austin, TX, Apr., 2022, pp. 1-6.
- [9]. **T. Bao**, H. Wang, W.-J. Wang, H.-C. Yang, and M. O. Hasna. “On the DOR and ADEP Analysis of RIS-aided URLLC Systems with Partial CSI in Smart Factory.” *IEEE GlobalCom*, Madrid, Spain, Dec., 2021, pp. 1-6.
- [10]. **T. Bao**, H.-C. Yang, and M. O. Hasna. “On the Ergodic Secrecy Rate of Massive MIMO Transmission with Partial Legitimate User CSI.” *IEEE PacRim*, Victoria, BC, Aug., 2019, pp. 1-5. (**Best Paper Award**)
- [11]. **T. Bao**, H.-C. Yang, and M. O. Hasna. “Secrecy Outage Performance Analysis of Massive MIMO Transmission with Multiple Non-Colluding Eavesdroppers and Partial Legitimate User CSI.” *IEEE VTC-Fall*, HI, Sep., 2019, pp. 1-5.
- [12]. **T. Bao**, H.-C. Yang, and M. O. Hasna. “Security Performance Analysis of MISOSE Transmission with Random Unitary Beamforming.” *IEEE VTC-Fall*, Chicago, IL, Aug. 2018, pp. 1-5.
- [13]. A. A. Gebremariam, **T. Bao**, D. Siracusa, et al. “Dynamic strict fractional frequency reuse for software-defined 5G networks.” *IEEE ICC*, Kuala Lumpur, Malaysia, May 2016, pp. 1-6.

- Patents

- [1]. **T. Bao**, M. Elsayed, P. E Iturria-Rivera, Y. Ozcant, M. Bavand, M. Erol-Kantarci. “Providing a Configuration of One or More Reconfigurable Intelligent Surfaces, and Training a Generative Inference Engine.” WIPO/PCT, IB 2025/058203 A1, Aug. 12, 2025.
- [2]. **T. Bao**, H.Wang, J.Wang, “Overhead reduction in RIS-aided URLLC systems”, U.S. Patent Application No. 18/705,230, Jan. 09, 2025.
- [3]. B. Liu, J. Wang, H. Wang, **T. Bao**, “Devices, methods, and computer readable storage medium for communication”, WIPO/PCT, WO 2024/093057 A1, May 10, 2024.
- [4]. **T. Bao**, L. Yang, J. Wang, T. Tang, H. Wang, “Secure transmission in RIS-assisted ISAC system”, WIPO/PCT, WO 2024/082583 A1, Apr. 25, 2024.
- [5]. **T. Bao**, L. Yang, J. Wang, H. Wang, “Handover management in RIS-assisted wireless networks”, WIPO/PCT, WO 2024/077596 A1, Apr. 18, 2024.
- [6]. L. Yang, **T. Bao**, J. Wang, T. Tang, H. Wang, “Methods and apparatuses for reconfigurable intelligent surface assisted positioning”, WIPO/PCT, WO 2024/082526 A1, Apr. 15, 2024.
- [7]. **T. Bao**, L. Yang, J. Wang, H. Wang, “ARIS location selection under ARIS-assisted wireless networks”, WIPO/PCT, WO 2024/074030 A1, Apr. 11, 2024.
- [8]. **T. Bao**, L. Yang, J. Wang, H. Wang, “Power control scheme for active RIS based on network requirements and RRC connection states”, WIPO/PCT, WO 2024/011577 A1, Jan. 11, 2024.
- [9]. J. Wang, B. Liu, C. Zhang, **T. Bao**, X. Guo, “Methods and apparatus of monitoring artificial intelligence model in radio access network”, WIPO/PCT, WO 2024/000559 A1, Jan. 4, 2024.
- [10]. B. Liu, J. Wang, C. Zhang, **T. Bao**, X. Guo, “UE initiated beam measurement and beam reporting”, WIPO/PCT, WO 2023/240601 A1, Dec. 21, 2023.
- [11]. J. Wang, B. Liu, C. Zhang, **T. Bao**, X. Guo, “CSI reporting methods and apparatus for AI enabled beam management”, WIPO/PCT, WO 2023/236193 A1, Dec. 21, 2023.
- [12]. J. Wang, **T. Bao**, H. Wang, “Methods and apparatus for uplink signal enhancement”, WIPO/PCT, WO 2023/221094 A1, Nov. 23, 2023.
- [13]. **T. Bao**, J. Wang, T. Tang, H. Wang, “Smart elements selection with NWDAF in RIS-aided URLLC Systems”, WIPO/PCT, WO 2023/108520 A1, Jun. 22, 2023.